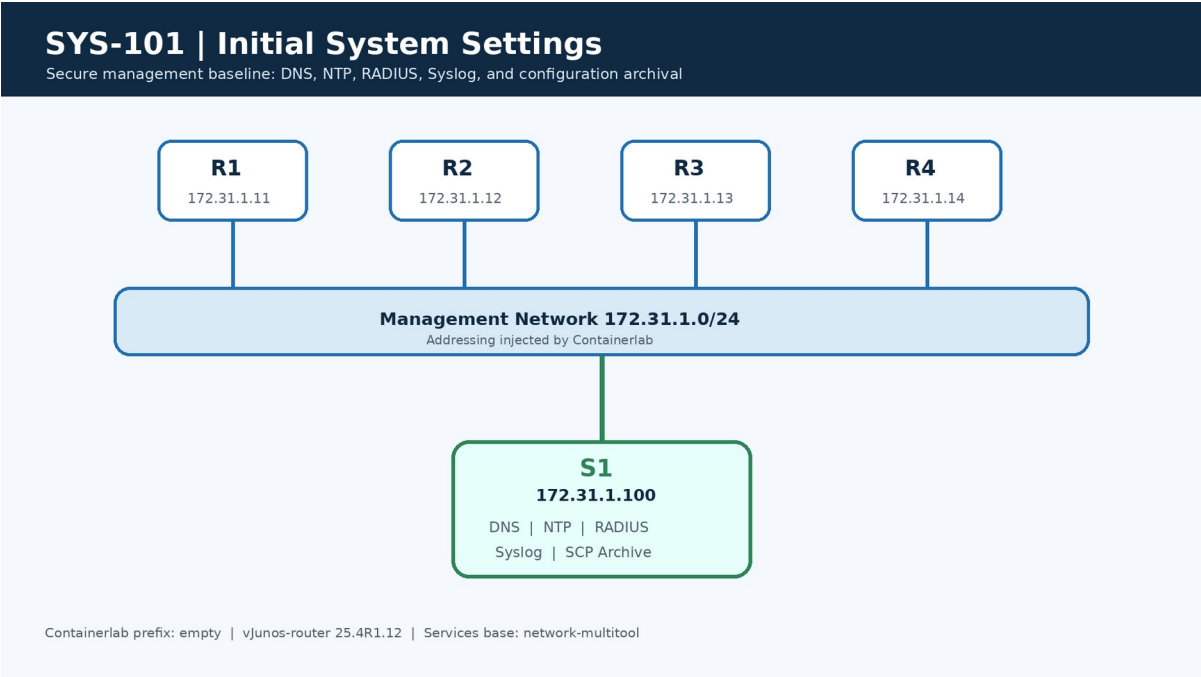


JNCIE-SP MODULAR LAB

SYS-101

Initial System Settings

Task-Only Assessment



PLATFORM	DURATION	TARGET
vJunos-router 25.4	3 hours	2 repetitions

Task-Only Assessment — Revision 3

Platform: Containerlab with four vJunos-router nodes and one services node

Time limit: 3 hours

Target repetitions: 2

Rules

- Do not open the solution guide before completing the first attempt.
- Preserve management access throughout the exercise.
- Use **commit confirmed** before AAA-sensitive changes.
- Record the requested evidence under **evidence/**.
- Do not reuse the lab credentials outside this isolated environment.

Addressing

Node	Containerlab management IP	Loopback	Required hostname	Role
R1	172.31.1.11/24	192.0.2.11/32	R1	Junos router
R2	172.31.1.12/24	192.0.2.12/32	R2	Junos router
R3	172.31.1.13/24	192.0.2.13/32	R3	Junos router
R4	172.31.1.14/24	192.0.2.14/32	R4	Junos router
S1	172.31.1.100/24	—	S1	DNS, NTP, RADIUS, Syslog and SCP archive

- > Containerlab/vrnetlab injects management addressing, **management-instance**,
- > **mgmt_junos**, and its default route. Do not configure them in startup files.
- > The vJunos management interface visible inside each VM is **fxp0.0** at
- > **10.0.0.15/24**, with management next hop **10.0.0.2**.

Task 1 - Validate baseline and identity

Objective

Confirm startup configuration loading, injected management connectivity, S1 reachability, and production identity.

Requirements

1. Deploy all five nodes and verify that they are operational.
2. Confirm Containerlab management addresses and the injected `fxp0.0` configuration.
3. Change bootstrap hostnames to R1, R2, R3, and R4.
4. Save deployment and management-interface evidence.

Task 2 - Central and local Syslog

Objective

Create troubleshooting evidence before implementing the remaining tasks.

Requirements

5. `jncie-sp-messages` must record information and interactive commands.
6. `user-commands` must record interactive commands.
7. S1 must receive notice messages and configuration changes through `mgmt_junos`.
8. User `ops` must receive warning messages when that user is logged in.
9. Archive three local files with a size of 100 KB each.

Task 3 - Management services and retained route

Objective

Enable required management protocols and preserve a remote management route if RPD is unavailable.

Requirements

10. Enable SSH, Telnet, and FTP.
11. Install `10.10.10.0/24` through `10.0.0.2` in `mgmt_junos`.
12. Retain the route if RPD stops.
13. Prevent the route from being advertised by dynamic routing protocols.

Task 4 - DNS and time zone

Objective

Provide consistent name resolution and administrative time display.

Requirements

14. Use S1 as the DNS server through `mgmt_junos`.
15. Use domain `sys101.lab`.
16. Set `Europe/Amsterdam` as the time zone.
17. Resolve and reach `s1.sys101.lab` through `mgmt_junos`.

Task 5 - Authenticated NTP

Objective

Synchronize every router with S1 using authenticated NTP.

Requirements

18. Configure the node-specific loopback address shown in the addressing table.
19. Use S1 as the operational NTP server through `mgmt_junos`.
20. Use MD5 key ID 1 and key value `workbook`.
21. Trust key ID 1 and associate it with S1.
22. Demonstrate an authenticated, reachable NTP association and synchronized status.

Task 6 - Configuration archival over SCP

Objective

Archive configuration automatically after every commit.

Requirements

23. Use S1 SCP with `lab / lab123` through `mgmt_junos`.
24. Store files in `/srv/ftp/archive` on S1.
25. Configure transfer on commit.
26. Trigger at least one post-configuration archive.
27. Demonstrate a non-empty archived configuration on S1.

Task 7 - Authorization classes and RADIUS mappings

Objective

Map RADIUS identities to distinct Junos authorization profiles.

Requirements

28. **ops-class** must have clear, network, reset, trace, and view permissions.
29. **noc-class** must have all permissions but deny clear, configure, edit, and start-shell commands.
30. Create local passwordless templates **radius-ops** and **radius-noc** for the two classes.
31. Create local passwordless template **remote** with read-only authorization.
32. Map RADIUS users **ops** and **noc** to their corresponding local templates with **Juniper-Local-User-Name**.
33. Demonstrate permitted and denied actions for each constrained role and read-only authorization for an unmapped RADIUS user.

Task 8 - RADIUS authentication with local fallback

Objective

Use centralized authentication first while maintaining local recovery access.

Requirements

34. Authentication order must be RADIUS followed by local password.
35. Use S1 through **mgmt_junos** with shared secret **workbook**.
36. Set retry to 1 and timeout to 2 seconds.
37. Preserve local break-glass user **lab / lab123**.
38. Prove RADIUS authentication while S1 is available.
39. Prove local fallback while S1 RADIUS is unavailable without causing administrative lockout.

Final acceptance

Run:

```
bash scripts/verify.sh
```

The last line must be **RESULT: PASS**. Preserve all evidence and record the attempt in the JNCIE-SP progress tracker.